Group 44
EARLY CANCER DETECTION

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FOR IMMEDIATE RELEASE

Group 44 Introduces Breakthrough miRNA-Based Diagnostic Tool for Breast Cancer Stage Classification

A new diagnostic system designed to give physicians clearer and more reliable guidance for staging breast cancer

Boca Raton, FL – September 9, 2025 – Group 44 today announced the upcoming release of a diagnostic system designed to improve the accuracy of breast cancer stage classification. By analyzing microRNA expression profiles, the molecular fingerprints that reveal how the disease develops, the system uncovers key patterns of its progression and translates them into clear staging guidance for clinicians. This research-based approach reduces diagnostic uncertainty, supports more confident treatment decisions, and reflects the growing role of data-driven tools in advancing patient care.

Our system reduces the uncertainty of breast cancer staging by analyzing microRNA signals and turning them into clear, usable answers for doctors. This means doctors get more accurate information, patients feel less anxious, and families gain peace of mind. Designed to fit seamlessly into hospital workflows, it provides reliable guidance that makes a difficult journey clearer and more supported.

Key Features:

- Advanced miRNA expression analysis using standardized TCGA datasets for breast cancer stage classification.
- Comparative evaluation of traditional machine learning (SVM, Random Forest) and deep learning approaches (MLP, 1D CNN).
- Rigorous 70/15/15 stratified data splitting protocol preventing bias and ensuring reliable model evaluation.
- Comprehensive feature selection comparison using Sequential Feature Selector, Variance Threshold, Chi-squared, and LASSO regularization methods.
- Statistical significance testing framework validating genuine performance improvements between methodologies.

Availability and Research Applications:

- Research prototype available for academic institutions and clinical research partnerships.
- Compatible with standard TCGA data processing workflows and university computing resources.
- Comprehensive documentation and reproducible methodology supporting peer review and validation.
- Integration capabilities with existing hospital information systems for clinical research applications.

About Group 44

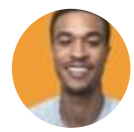
Group 44 is a student team at Florida Atlantic University dedicated to creating solutions that make healthcare clearer and more supportive. Our Breast Cancer Stage Classification System combines advanced data science with compassion, turning complex microRNA signals into clear answers that doctors and patients can trust. At the heart of our work is a simple belief: innovation matters most when it helps people feel informed, supported, and hopeful.

Meet the Team

Joshua Adegboyo, Team Lead

Anika Baro Villa

Nicholas Campos



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